

DAY 00 / MASTER INDEX

Week 1 Notes Collection

Topic Index

Seven Markdown source notes and their validated companion PDFs in Notes/Week-1.

7

TOPIC NOTES

371

PDF PAGES

19,454

SOURCE LINES

RECOMMENDED PATH

1

Python Backend Foundations and Production API Contracts

Days 01–02

2

Databases, Concurrency, and Experimentation

Days 03–05

3

Finance Analytics Service PoC and Week 1 Consolidation

Days 06–07

HOW TO USE THIS INDEX Open the matching Markdown source for source content or the companion PDF for reading and review.

MODULE 01 / DAYS 01–02

Python Backend Foundations and Production API Contracts

Exact source headings, filenames, and validated companion-PDF page counts.

DAY

01

Day 1 — Role Baseline and Python Backend Foundations for Applied AI

MAJOR SECTIONS 1. Senior Applied AI/ML competency model; 2. Mapping your resume to the competency model; 3. What backend engineering means in an applied AI platform; 4. How Python is used across the AI lifecycle; 5. Recommended Python structure for an AI service

SOURCE [Day-1-role-diagnostic-python-backend-architecture-clean-code-arrays.md](#) | PDF 54 pages

DAY

02

Day 2 — Production API Contracts for Applied AI Systems

MAJOR SECTIONS 1. REST, streaming, asynchronous jobs, and gRPC; 2. High-level architecture; 3. Recommended FastAPI application structure; 4. Pydantic request and response contracts; 5. Status codes and consistent errors

SOURCE [Day-2-fastapi-rest-pydantic-api-contracts-versioning-and-idempotency-strings.md](#) | PDF 64 pages

MODULE 02 / DAYS 03–05

Databases, Concurrency, and Experimentation

Exact source headings, filenames, and validated companion-PDF page counts.

DAY

03

Day 3 — Databases, Analytical SQL, and Caching for Applied AI Systems

MAJOR SECTIONS 1. Main alternatives and selection criteria; 2. Reference architecture; 3. SQL versus NoSQL at senior interview depth; 4. Relational design principles; 5. Core production schemas

SOURCE [Day-3-postgresql-analytical-sql-nosql-redis-and-caching-hashmap.md](#) | PDF 71 pages

DAY

04

Day 4 — Production Python Concurrency, Testing, Observability, and Resilience

MAJOR SECTIONS 1. Choosing the concurrency model; 2. Synchronous versus asynchronous execution; 3. Understanding the event loop; 4. `gather`, semaphores, queues, and bounded concurrency; 5. Architecture

SOURCE [Day-4-async-python-concurrency-retries-testing-logging-and-debugging-two-pointers.md](#) | PDF 50 pages

DAY

05

Day 5 — Statistics, Experimentation, and Causal Reasoning

MAJOR SECTIONS 1. Descriptive statistics and distributions; 2. Conditional probability, Bayes, expected value, and uncertainty; 3. Sampling and bias; 4. Confidence intervals; 5. Hypothesis testing

SOURCE [Day-5-statistics-probability-eda-and-experimentation-foundations-sliding-window.md](#) | PDF 51 pages

MODULE 03 / DAYS 06–07

Finance Analytics Service PoC and Week 1 Consolidation

Exact source headings, filenames, and validated companion-PDF page counts.

DAY

06

Day 6 — Finance Analytics Service PoC

MAJOR SECTIONS Problem statement; Completed deliverable; Architecture; Executed results; Main documents

SOURCE Day-6-weekend-poc-1-finance-analytics-api-with-sql-statistics-and-caching-stack.md | PDF 48 pages

DAY

07

Day 7 — Week 1 Consolidation and Strict Review

MAJOR SECTIONS Revision summary; Priority table; Part A — Closed-book assessment; Stop here until the assessment is complete; Part B — Strict review and answer key

SOURCE Day-7-week-1-revision-backend-analytics-mock-and-poc-review-queue.md | PDF 33 pages

QUICK LOCATOR

Find a topic fast

Source title and first substantive section for each note.

Source title	Start with	Best note
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Day 2 — Production API Contracts for Applied AI Systems	1. REST, streaming, asynchronous jobs, and gRPC	Day 02
Day 3 — Databases, Analytical SQL, and Caching for Applied AI Systems	1. Main alternatives and selection criteria	Day 03
Day 4 — Production Python Concurrency, Testing, Observability, and Resilience	1. Choosing the concurrency model	Day 04
Day 5 — Statistics, Experimentation, and Causal Reasoning	1. Descriptive statistics and distributions	Day 05
Day 6 — Finance Analytics Service PoC	Problem statement	Day 06
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COLLECTION SUMMARY

Source format	7 Markdown files in Notes/Week-1/
Reading format	7 companion note PDFs in Notes/Week-1/pdfs/
Total coverage	19,454 source lines and 371 note-PDF pages
Index generated	06 August 2026